

A Motivating example:

$$\sum_{n=1}^{\infty} \frac{1}{2^n+1} \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{2^n}$$

The Second Series is a geometric Series that converges to 1.

The first Series is similar to the first
So we think it may also converge.

We notice $0 \leq \frac{1}{2^n+1} < \frac{1}{2^n}$ for all n

We then have

$$0 \leq \sum \frac{1}{2^n+1} < \sum \frac{1}{2^n} = 1$$

So $\sum \frac{1}{2^n+1}$ converges by comparison.

Comparison Test

Suppose that $\sum a_n$ and $\sum b_n$ are series of positive terms,

1) If $\sum b_n$ is convergent and $\underline{a_n \leq b_n}$ for all n , then $\sum a_n$ is also convergent.

2) If $\sum b_n$ is divergent and $\underline{a_n \leq b_n}$ for all n , then $\sum a_n$ is also divergent.

Pf/ Let $S_n = \sum_{i=1}^n a_i$, $t_n = \sum_{i=1}^n b_i$, $t = \sum_{n=1}^{\infty} b_n$

Since both series are of pos.ve terms, the sequences $\{S_n\}$ and $\{t_n\}$ are increasing ($S_{n+1} = S_n + a_{n+1} \geq S_n$). Also $t_n \rightarrow t$, so $t_n \leq t$ for all n . Since $a_i \leq b_i$, we have $S_n \leq t_n$ for all n . So, $\{S_n\}$ is increasing and bounded above and therefore converges by Monotone Convergence Thm. Thus $\sum a_n$ converges.

If $\sum b_n$ diverges, $t_n \rightarrow \infty$ (since $\{t_n\}$ increases). But $a_i \leq b_i$ so

So $S_n \geq tn$. Thus $S_n \rightarrow \infty$, therefore $\sum a_n$ diverges.

In practice we compare to either

• A p-series $\sum \frac{1}{n^p}$, or

• A geometric series $\sum ar^n$

Ex:

$$\sum \frac{5}{2n^2 + 4n + 3} \quad \text{Converges}$$

$$\frac{5}{2n^2 + 4n + 3} < \frac{5}{2n^2}$$

$$\sum \frac{5}{2n^2} = \frac{5}{2} \sum \frac{1}{n^2} \quad \text{Converges}$$

$$\text{So } \sum \frac{5}{2n^2 + 4n + 3} \quad \text{Converges}$$

Ex: $\sum_{n=1}^{\infty} \frac{\ln n}{n}$ diverges

We showed this via Integral test before. But now
we notice

$$\frac{\ln(n)}{n} > \frac{1}{n}$$

and

$$\sum \frac{1}{n} \text{ diverges (P-series w/ } p=1)$$

$$\therefore \sum \frac{\ln(n)}{n} \text{ diverges too.}$$

The Comparison test is useful, but we need the inequality to be correct to use it.

Consider

$$\sum \frac{1}{2^n - 1}$$

This looks like $\sum \frac{1}{2^n}$ so we think it converges.

But, we can't use normal comparison test as

$$\frac{1}{2^n - 1} > \frac{1}{2^n}$$

To evaluate the sum we use

Limit Comparison Test

Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms. If

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c$$

where c is a finite number and $c > 0$, then

- Both $\sum a_n$ and $\sum b_n$ converge, or
- Both $\sum a_n$ and $\sum b_n$ diverge

Pf/ Let m and M be such that

$$m < C < M, \quad \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = C \quad \text{so there is}$$

an integer N s.t.

$$m < \frac{a_n}{b_n} < M \quad \text{when } n > N$$

So

$$mb_n < a_n < Mb_n \quad \text{when } n > N$$

If $\sum b_n$ converges, so does $\sum Mb_n$. Thus $\sum a_n$ converges by Comparison test

If $\sum b_n$ diverges, so does $\sum mb_n$. So $\sum a_n$ diverges by Comparison test.

Ex: $\sum_{n=1}^{\infty} \frac{1}{2^n - 1}$

we compare to $\sum \frac{1}{2^n}$

$$a_n = \frac{1}{2^n - 1}, \quad b_n = \frac{1}{2^n}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\frac{1}{2^n - 1}}{\frac{1}{2^n}} = \lim_{n \rightarrow \infty} \frac{2^n}{2^n - 1} = \lim_{n \rightarrow \infty} \frac{1}{1 - \frac{1}{2^n}} = 1$$

So $\sum \frac{1}{2^n - 1}$ converges

Ex:

$$\sum_{n=1}^{\infty} \frac{2n^2 + 3n}{\sqrt{5+n^5}}$$

$$a_n = \frac{2n^2 + 3n}{\sqrt{5+n^5}}, \quad b_n = \frac{2n^2}{n^{5/2}} = \frac{2}{n^{1/2}}$$

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{a_n}{b_n} &= \frac{\frac{2n^2+3n}{\sqrt{5+n^5}}}{\frac{2}{n^{1/2}}} = \lim_{n \rightarrow \infty} \frac{2n^2+3n}{\sqrt{5+n^5}} \cdot \frac{n^{1/2}}{2} \\ &= \lim_{n \rightarrow \infty} \frac{2n^{5/2} + 3n^{3/2}}{2\sqrt{5+n^5}} \cdot \frac{\frac{1}{n^{1/2}}}{\frac{1}{n^{1/2}}} \\ &= \lim_{n \rightarrow \infty} \frac{2 + \frac{3}{n}}{2\sqrt{\frac{5}{n^5} + 1}} \\ &= \frac{2}{2} = 1 \end{aligned}$$

Since $\sum b_n = \sum \frac{2}{n^{1/2}}$ diverges (P-Series $p = \frac{1}{2}$)

$$S. \sum a_n = \sum \frac{2n^2 + 3n}{\sqrt{5+n^5}} \quad \boxed{\text{diverges}}$$

Note, when finding a series to compare to, we usually take the highest order term in numerator and denominator.